

Zepto SQL Analysis

A SQL-Based Data Cleaning & Business Analysis Project

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Tools: PostgreSQL

Repository: [github.com/<shalini090605>/zepto-sql-analysis](https://github.com/shalini090605/zepto-sql-analysis)

1. Introduction

Zepto is a quick-commerce (fast grocery delivery) platform. This project analyzes a snapshot of Zepto's product catalog using SQL, simulating the kind of work a data or business analyst would do for a retail/e-commerce team: taking a raw, messy product table and turning it into clean, decision-ready answers about pricing, discounts, stock availability, and inventory.

The goal was not just to write queries, but to follow a realistic analyst workflow — explore the data first, identify and fix data-quality issues, and only then answer specific business questions with confidence that the underlying numbers are trustworthy.

2. Objective

- Explore a raw product dataset to understand its structure and spot data-quality issues.
- Clean the data (handle invalid prices, fix unit inconsistencies) before analysis.
- Answer 8 realistic business questions covering pricing strategy, stock management, revenue estimation, and inventory.
- Demonstrate core SQL skills: aggregation, filtering, grouping, CASE logic, and sorting.

3. Dataset Overview

The dataset is stored in a single table, `zepto`, with the following structure:

Column	Data Type	Description
<code>sku_id</code>	SERIAL (PK)	Unique product identifier
<code>category</code>	VARCHAR(120)	Product category
<code>name</code>	VARCHAR(200)	Product name
<code>mrp</code>	NUMERIC(8,2)	Maximum Retail Price
<code>discountPercent</code>	NUMERIC(5,2)	Discount offered on MRP
<code>availableQuantity</code>	INTEGER	Units available in stock
<code>discountedSellingPrice</code>	NUMERIC(8,2)	Final price after discount
<code>weightInGms</code>	INTEGER	Product weight in grams
<code>outOfStock</code>	BOOLEAN	Whether the product is out of stock
<code>quantity</code>	INTEGER	Order/package quantity

4. Methodology

4.1 Table Creation

A zepto table was created in PostgreSQL with appropriate data types and a primary key (sku_id) to uniquely identify each product record.

4.2 Data Exploration

Before touching the data, the following checks were run to understand its shape and quality:

- Total row count, to know the overall size of the dataset.
- A sample of 10 rows, to visually inspect the data.
- A combined NULL check across all key columns (name, category, price, discount, stock, weight, quantity fields), to catch incomplete records in one query.
- Distinct product categories, to understand the scope of the catalog.
- In-stock vs. out-of-stock product counts.
- Products appearing under the same name multiple times (multiple SKUs — e.g. different pack sizes of the same product).

4.3 Data Cleaning

Two data-quality issues were identified and fixed before analysis:

- Invalid pricing: Some rows had mrp or discountedSellingPrice equal to 0, which is not a realistic price. Rows with mrp = 0 were removed, since a product with no listed price cannot be meaningfully analyzed.
- Unit inconsistency: Prices were stored in paise rather than rupees. Both mrp and discountedSellingPrice were divided by 100 to convert them into a consistent, human-readable rupee format.

Cleaning the data before analysis matters because unnoticed issues like these would silently distort every downstream number — for example, a revenue estimate calculated in paise instead of rupees would be off by a factor of 100.

5. Business Questions & Analytical Approach

Eight business questions were answered using aggregation, filtering, and conditional logic in SQL:

Q1. Top 10 best-value products by discount percentage

```
select distinct name,mrp,discountPercent
```

```
from zepto
```

```
order by discountPercent desc
```

```
limit 10;
```

	name character varying (200)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dukes Waffy Orange Wafers	45.00	51.00
2	Dukes Waffy Chocolate Wafers	45.00	51.00
3	Dukes Waffy Strawberry Wafers	45.00	51.00
4	RRO Mozzarella Block Cheese	295.00	50.00
5	Ceres Foods Laal Maas Instant Liquid Mas...	220.00	50.00
6	Chef's Basket Durum Wheat Elbow Pasta	160.00	50.00

Sorted all products by discountPercent in descending order to surface the biggest markdowns — useful for a marketing team building a 'best deals' promotion.

Q2. High-MRP products that are out of stock

```
select name, mrp
```

```
from zepto
```

```
where outOfStock is True and mrp > 300
```

```
order by mrp desc;
```

	name character varying (200)	mrp numeric (8,2)
1	Patanjali Cow's Ghee	565.00
2	Patanjali Cow's Ghee	565.00
3	MamyPoko Pants Standard Diapers, Extra Large (12 - 17 ...	399.00
4	MamyPoko Pants Standard Diapers, Extra Large (12 - 17 ...	399.00
5	Aashirvaad Atta With Mutigrains	315.00

Filtered for outOfStock = true and mrp > ₹300, sorted by price — flags premium products at risk of lost revenue due to stockouts, which operations/inventory teams would want to restock first.

Q3. Estimated revenue per category

```
select category,
```

```
sum(discountedSellingPrice*availableQuantity) as total_revenue
```

```
from zepto
```

```
group by category
```

```
order by total_revenue;
```

	category character varying (120)	total_revenue numeric
1	Fruits & Vegetables	10846.00
2	Meats, Fish & Eggs	20693.00
3	Biscuits	25007.60
4	Beverages	55051.00
5	Dairy, Bread & Batter	55051.00

Multiplied discountedSellingPrice by availableQuantity and summed by category — a proxy for potential revenue if all current stock sold at the discounted price, helping identify which categories drive the most value.

Q4. Premium products with a weak discount (MRP > ₹500, discount < 10%)

```
select distinct name,mrp,discountPercent
```

```
from zepto
```

```
where mrp>500 and discountPercent<10
```

```
order by mrp,discountPercent desc;
```

	name character varying (200)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Amul Pure Ghee (Pouch)	505.00	1.00
2	Fresh Mutton Mince - Kheema (Halal)	515.00	3.00
3	Amul Ghee (Tin)	520.00	2.00
4	Surf Excel Matic Front Load Detergent Powder	530.00	4.00
5	Kellogg's Real Almond & Honey Corn Flakes	550.00	5.00

Filtered for high-price, low-discount combinations — useful for a pricing team evaluating whether premium products are being discounted competitively enough.

Q5. Top 5 categories by average discount percentage

Grouped by category and averaged discountPercent, then ranked — shows which categories are being discounted most aggressively overall, which could reflect competitive pressure or excess stock.

```
select category,round(avg(discountPercent),2) as avg_discount
```

```
from zepto
```

```
group by category
```

```
order by avg_discount desc
```

```
limit 5;
```

	category character varying (120) 🔒	avg_discount numeric 🔒
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32
4	Chocolates & Candies	8.32
5	Packaged Food	8.32

Q6. Price per gram for products above 100g

```
select distinct name,weightInGms,discountedSellingPrice,
round(discountedSellingPrice/weightInGms,2) as price_per_gram
from zepto
where weightInGms>=100
order by price_per_gram;
```

	name character varying (200) 🔒	weightInGms integer 🔒	discountedSellingPrice numeric (8,2) 🔒	price_per_gram numeric 🔒
1	Tata Salt	1000	24.00	0.02
2	Onion	1000	21.00	0.02
3	Onion	3000	57.00	0.02
4	Shubh kart - Nirmal sugandhi mogra wet dhoop zipper 20 sticks	1160	28.00	0.02
5	Vicks Cough Drops Menthol	1160	20.00	0.02

Calculated $\text{discountedSellingPrice} \div \text{weightInGms}$ for products weighing 100g or more, then sorted ascending — surfaces the best value-for-weight products, similar to a 'unit price' comparison shoppers use.

Q7. Weight-based segmentation (Low / Medium / Bulk)

```
select distinct name,weightInGms,
case when weightInGms<1000 then 'Low'
      when weightInGms<5000 then 'Medium'
      else 'Bulk'
end as weight_category
from zepto;
```

	name character varying (200)	weightInGms integer	weight_category text
1	Popular Essentials Poppy Seeds (Gasagase)	100	Low
2	Origami Kitchen Towels 2 In 1 Quilted 2ply	116	Low
3	Origami Good Karma Paper Serviettes - 100 napkins (1 ply)	58	Low
4	Johnson's Baby Top To Toe Bath Body Wash	210	Low
5	Harpic Hygienic Toilet Cleaner Rim Block Citrus	26	Low

Used a CASE statement to bucket products into Low (<1,000g), Medium (1,000–4,999g), and Bulk (5,000g+) — turns a continuous number into a simple category useful for logistics and packaging decisions.

Q8. Total inventory weight per category

select category,

sum(weightInGms*availableQuantity) as total_weight

from zepto

group by category

order by total_weight;

	category character varying (120)	total_weight bigint
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735

Multiplied weightInGms by availableQuantity and summed by category — estimates the physical warehouse/storage load each category represents, relevant for space planning and logistics.

6. Key Skills Demonstrated

- Database design: table creation with appropriate data types and a primary key.
- Data quality checks: systematic NULL detection and handling.
- Data cleaning: identifying and fixing invalid values and unit inconsistencies.
- SQL querying: aggregation (SUM, AVG, COUNT), filtering (WHERE), grouping (GROUP BY, HAVING), conditional logic (CASE), and sorting (ORDER BY).
- Business framing: translating raw queries into questions a retail/operations team would actually ask.

7. Conclusion

This project shows a complete, realistic analyst workflow on e-commerce data — starting from raw, unclean data and ending with specific, actionable answers to business questions around pricing, stock, and inventory. The full SQL script and documentation are available in the accompanying GitHub repository.

Note: this report describes the structure and logic of each query. To include specific result values (e.g. exact revenue figures or top products), run `zepto_sql_analysis.sql` against the full dataset and add the output tables or screenshots here.